

Structured Dynamic Panel Conditioning for KDN-2: An Exact Chunkwise Algorithm with Dense Boundary Covariance

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Abstract

Exact Kalman linear attention carries a dense posterior covariance but pays a sequential dense covariance dependency at every token. A natural approximation is to hold the latent state fixed inside a short panel, condition all panel observations jointly, and add the accumulated process noise only at the boundary. That approximation admits compact panel solves and is internally consistent, but a sealed quality study rejects it: when the transition strongly contracts, the observations primarily measure newly injected process noise, precisely the information that the delayed-noise model discards.

This note develops a different panel algorithm. It retains the process-noise contribution to the observation covariance, to every causal read–observation cross-covariance, and to the terminal state–observation cross-covariance. The resulting panel update is algebraically identical to the tokenwise Kalman filter. For the damped rotational process model of KDN-2 and block-isotropic process noise, these dynamic terms can be constructed in $O(b^2d)$ work for panel width b and state dimension d , alongside the unavoidable $O(bd^2)$ dense-boundary work. Thus panel width is an execution tile rather than a statistical approximation knob. Performance remains an empirical kernel question; this document defines the mathematics, a coherent nested fallback approximation, and the gates required before making a speed claim.

1 Decision and scope

The exact-static, panel-refreshed proposal is not promoted as “ChunkKLA v1.” Its Cholesky conditioning is exact for a static latent variable, but relative to the tokenwise dynamic model it sets all within-panel process-noise observation correlations to zero and adds only the open-loop terminal covariance. In a preregistered FP64 study, width $b = 1$ was an exact control, while every tested width $b \geq 2$ failed at least one regime. At $b = 2$, the near-zero-transition regime had median output relative error 0.69719 and median final-mean relative error 0.99999. This is a modeling failure rather than a factorization or roundoff failure.

The replacement is *structured dynamic panel conditioning*. It preserves

1. the token-timed damped rotational transition;
2. every process-noise contribution that can be inferred from a panel observation;
3. causal read-after-write outputs;
4. the exact dense posterior covariance at every panel boundary; and
5. the tokenwise KDN-2 result in exact arithmetic, for every panel width.

KDN-2 is the model name. Panel conditioning is an implementation strategy, not a second model. The old exact-static construction is retained only as the *delayed-noise ablation*. Production files remain out of scope until a sandbox implementation passes forward, gradient, finite-precision, resource, and end-to-end speed gates.

2 Tokenwise KDN-2

Let $\mathbf{S}_t^* \in \mathbb{R}^{d \times d_v}$ be the latent associative state and $\mathbf{v}_t, \mathbf{e}_t, \mathbf{o}_t \in \mathbb{R}^{d_v}$. Conditional on the incoming filtration, its value columns are independent with a shared address-space covariance,

$$\mathbf{S}_{0,:,a}^* \mid \mathcal{F}_0 \sim \mathcal{N}(\mathbf{S}_{0,:,a}, \mathbf{P}_0), \quad \mathbf{P}_0 \succeq \mathbf{0}, \quad a = 1, \dots, d_v. \quad (1)$$

The latent process and observation are

$$\mathbf{S}_t^* = \mathbf{D}_t \mathbf{S}_{t-1}^* + \mathbf{W}_t, \quad \mathbf{W}_{t,:,a} \sim \mathcal{N}(\mathbf{0}, \mathbf{\Omega}_t), \quad (2)$$

$$\mathbf{v}_t = \mathbf{S}_t^{*\top} \mathbf{k}_t + \mathbf{e}_t, \quad \mathbf{e}_t \sim \mathcal{N}(\mathbf{0}, r_t \mathbf{I}_{d_v}), \quad r_t > 0. \quad (3)$$

Every $\mathbf{\Omega}_t$ is symmetric positive semidefinite. The process noises, observation noises, incoming state error, and different value-coordinate noise columns are mutually independent. All covariance and conditioning identities below are derived for one arbitrary value column and then applied to all d_v columns with shared factors. The transition, keys, queries, and noise parameters are conditioned design quantities known causally at their token. Value coordinates are conditionally independent and share the same address-space covariance. To connect with the lifted Direction-A formulation, the symbols used below denote the effective state-space addresses

$$\mathbf{k}_t = \mathbf{E}_t \mathbf{k}_t^{\text{raw}}, \quad \mathbf{q}_t = \mathbf{F}_t \mathbf{q}_t^{\text{raw}}, \quad (4)$$

where the lifts $\mathbf{E}_t$ and $\mathbf{F}_t$ are also token-causal design quantities.

The tokenwise filter is

$$\hat{\mathbf{S}}_t = \mathbf{D}_t \mathbf{S}_{t-1}, \quad \hat{\mathbf{P}}_t = \mathbf{D}_t \mathbf{P}_{t-1} \mathbf{D}_t^\top + \mathbf{\Omega}_t, \quad (5)$$

$$\kappa_t = \frac{\hat{\mathbf{P}}_t \mathbf{k}_t}{r_t + \mathbf{k}_t^\top \hat{\mathbf{P}}_t \mathbf{k}_t}, \quad \mathbf{S}_t = \hat{\mathbf{S}}_t + \kappa_t (\mathbf{v}_t - \hat{\mathbf{S}}_t^\top \mathbf{k}_t)^\top, \quad (6)$$

$$\mathbf{P}_t = \hat{\mathbf{P}}_t - \frac{(\hat{\mathbf{P}}_t \mathbf{k}_t)(\hat{\mathbf{P}}_t \mathbf{k}_t)^\top}{r_t + \mathbf{k}_t^\top \hat{\mathbf{P}}_t \mathbf{k}_t}, \quad \mathbf{o}_t = \mathbf{S}_t^\top \mathbf{q}_t. \quad (7)$$

The covariance is dense. The challenge is not the formula but the tokenwise chain of dense $d \times d$ updates.

2.1 Damped rotational process

Partition the address dimension into fixed modes of size $d_m = 2r_m$. KDN-2 uses

$$\mathbf{D}_t^{(m)} = \rho_{t,m} \mathbf{R}(\theta_{t,m}) \otimes \mathbf{I}_{r_m}, \quad \mathbf{D}_t = \text{blockdiag}_m \mathbf{D}_t^{(m)}, \quad (8)$$

where

$$\mathbf{R}(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}. \quad (9)$$

The radius controls survival, the angle controls phase, and the key/query geometry controls visibility. Process noise is block isotropic,

$$\mathbf{\Omega}_t = \text{blockdiag}_m(\nu_{t,m} \mathbf{I}_{d_m}), \quad \nu_{t,m} \geq 0. \quad (10)$$

This structure is a model choice. The covariance $\mathbf{P}_t$ itself is not projected: measurement updates create anisotropy and cross-mode correlations, and the panel boundary retains them. Ordinary learned transitions use $0 < \rho_{t,m} \leq 1$. A hard reset is an explicit override $\mathbf{D}_t = \mathbf{0}$ while retaining $\mathbf{\Omega}_t$; it may occur inside a mathematical panel. At such a token, every effective mode radius used by the covariance builder is zero even if the network emitted a nonzero pre-reset radius.

3 One exact dynamic panel

Let a panel contain local tokens $1:b$ and begin after posterior $(\mathbf{S}_0, \mathbf{P}_0)$. The first prediction occurs at its original time:

$$\mathbf{X}^- = \mathbf{D}_1 \mathbf{S}_0, \quad \mathbf{P}^- = \mathbf{D}_1 \mathbf{P}_0 \mathbf{D}_1^\top + \mathbf{\Omega}_1. \quad (11)$$

The token-one state is the common panel latent variable. This convention makes $b = 1$ exactly the tokenwise filter and leaves the later noises $\mathbf{w}_{2:b}$ to be handled dynamically. A partial final panel uses its actual token count as b ; padding is excluded from every covariance, factorization, solve, and output rather than represented as a finite-noise observation. An implementation may use a fixed physical tile only if inactive rows are algebraically block-decoupled from the true $b \times b$ system.

3.1 Deterministic transport and dynamic residual state

Define forward products

$$\mathbf{\Phi}_1 = \mathbf{I}, \quad \mathbf{\Phi}_i = \mathbf{D}_i \mathbf{D}_{i-1} \cdots \mathbf{D}_2 \quad (i \geq 2), \quad (12)$$

and transport the observation and read addresses into the token-one frame:

$$\tilde{\mathbf{k}}_i = \mathbf{\Phi}_i^\top \mathbf{k}_i, \quad \tilde{\mathbf{q}}_i = \mathbf{\Phi}_i^\top \mathbf{q}_i. \quad (13)$$

No inverse transition or prefix ratio is used.

Separate the later process noise from the transported entry state:

$$\mathbf{s}_i^* = \mathbf{\Phi}_i \mathbf{x}^* + \boldsymbol{\xi}_i, \quad \boldsymbol{\xi}_1 = \mathbf{0}, \quad \boldsymbol{\xi}_i = \mathbf{D}_i \boldsymbol{\xi}_{i-1} + \mathbf{w}_i. \quad (14)$$

Here $\mathbf{x}^*$ has conditional mean $\mathbf{X}^-$ and covariance $\mathbf{P}^-$. Let

$$\mathbf{Q}_1 = \mathbf{0}, \quad \mathbf{Q}_i = \mathbf{D}_i \mathbf{Q}_{i-1} \mathbf{D}_i^\top + \mathbf{\Omega}_i. \quad (15)$$

For $i \geq j$, define

$$\mathbf{\Phi}_{i:j} = \mathbf{D}_i \mathbf{D}_{i-1} \cdots \mathbf{D}_{j+1}, \quad \mathbf{\Phi}_{j:j} = \mathbf{I}. \quad (16)$$

Then

$$\mathbf{Q}_{ij}^W := \text{Cov}(\boldsymbol{\xi}_i, \boldsymbol{\xi}_j) = \mathbf{\Phi}_{i:j} \mathbf{Q}_j. \quad (17)$$

This identity remains valid for an exact-zero transition: earlier covariance is erased, and the current process noise starts a new independent component.

3.2 Joint innovation covariance

Stack

$$\widetilde{\mathbf{K}} = [\widetilde{\mathbf{k}}_1, \dots, \widetilde{\mathbf{k}}_b], \quad \widetilde{\mathbf{Q}} = [\widetilde{\mathbf{q}}_1, \dots, \widetilde{\mathbf{q}}_b], \quad (18)$$

and let row i of $\mathbf{V} \in \mathbb{R}^{b \times d_v}$ be $\mathbf{v}_i^\top$. With $\mathbf{R}_v = \text{diag}(r_1, \dots, r_b)$, the innovation matrix is

$$\Delta = \mathbf{V} - \widetilde{\mathbf{K}}^\top \mathbf{X}^-. \quad (19)$$

The later process noises induce the observation covariance

$$[\Sigma_W]_{ij} = \mathbf{k}_i^\top \mathbf{Q}_{ij}^W \mathbf{k}_j \quad (i \geq j), \quad [\Sigma_W]_{ji} = [\Sigma_W]_{ij}. \quad (20)$$

Therefore the exact innovation covariance is

$$\mathbf{G} = \mathbf{R}_v + \widetilde{\mathbf{K}}^\top \mathbf{P}^- \widetilde{\mathbf{K}} + \Sigma_W. \quad (21)$$

Because $\mathbf{R}_v \succ 0$, $\mathbf{G}$ is positive definite in exact arithmetic. Factor and whiten the innovations:

$$\mathbf{G} = \mathbf{J} \mathbf{J}^\top, \quad \mathbf{Z} = \mathbf{J}^{-1} \Delta, \quad (22)$$

where $\mathbf{J}$ is lower triangular.

3.3 All causal outputs

For $j \leq i$, the state–observation cross-covariance is

$$\mathbf{C}_{i,j} = \text{Cov}(\mathbf{s}_i^*, v_j) = \Phi_i \mathbf{P}^- \widetilde{\mathbf{k}}_j + \mathbf{Q}_{ij}^W \mathbf{k}_j. \quad (23)$$

Define a lower-triangular read–observation matrix

$$[\mathbf{H}]_{ij} = \begin{cases} \mathbf{q}_i^\top \mathbf{C}_{i,j}, & j \leq i, \\ 0, & j > i. \end{cases} \quad (24)$$

Every read-after-write output is then produced by

$$\boxed{\mathbf{O} = \widetilde{\mathbf{Q}}^\top \mathbf{X}^- + \text{tril}(\mathbf{H} \mathbf{J}^{-\top}) \mathbf{Z}.} \quad (25)$$

The triangular mask is essential. For row i , the first i columns of the full triangular solve equal the solve using the leading $i \times i$ Cholesky block. Hence output i is the conditional mean using observations $1:i$ and cannot depend on a future token. Masking a full $\mathbf{G}^{-1}$ update is not equivalent.

3.4 Dense boundary posterior

At the terminal token, collect

$$\mathbf{C}_b = [\mathbf{C}_{b,1}, \dots, \mathbf{C}_{b,b}], \quad \mathbf{B}_b = \mathbf{C}_b \mathbf{J}^{-\top}. \quad (26)$$

The exact posterior mean and covariance are

$$\boxed{\mathbf{S}_b = \Phi_b \mathbf{X}^- + \mathbf{B}_b \mathbf{Z},} \quad (27)$$

$$\boxed{\mathbf{P}_b = \Phi_b \mathbf{P}^- \Phi_b^\top + \mathbf{Q}_b - \mathbf{B}_b \mathbf{B}_b^\top} \quad (28)$$

The next panel consumes this dense $\mathbf{P}_b$ without projection. Because terminal process noise also appears in the observations, the ordinary independent-noise Joseph formula is not the right reference. Let

$$\mathbf{\Pi}_b = \Phi_b \mathbf{P}^- \Phi_b^\top + \mathbf{Q}_b, \quad \mathbf{L}_b = \mathbf{C}_b \mathbf{G}^{-1} = \mathbf{B}_b \mathbf{J}^{-1}. \quad (29)$$

The generalized estimator-error form

$$\mathbf{P}_b = \mathbf{\Pi}_b - \mathbf{L}_b \mathbf{C}_b^\top - \mathbf{C}_b \mathbf{L}_b^\top + \mathbf{L}_b \mathbf{G} \mathbf{L}_b^\top \quad (30)$$

is an independent FP64 numerical reference and reduces algebraically to equation (28). The direct downdate is admitted to an FP32 candidate only if the full-sequence numerical gate accepts it.

A stronger positive-semidefinite reference exposes the independent Gaussian sources. For $s = 2:b$, let $\mathbf{T}_s = \Phi_{b:s}$ and let $\mathbf{A}_s \in \mathbb{R}^{b \times d}$ have row

$$[\mathbf{A}_s]_{i,:} = \begin{cases} \mathbf{k}_i^\top \Phi_{i:s}, & i \geq s, \\ \mathbf{0}^\top, & i < s. \end{cases} \quad (31)$$

Then

$$\begin{aligned} \mathbf{P}_b &= (\Phi_b - \mathbf{L}_b \widetilde{\mathbf{K}}^\top) \mathbf{P}^- (\Phi_b - \mathbf{L}_b \widetilde{\mathbf{K}}^\top)^\top \\ &\quad + \sum_{s=2}^b (\mathbf{T}_s - \mathbf{L}_b \mathbf{A}_s) \mathbf{\Omega}_s (\mathbf{T}_s - \mathbf{L}_b \mathbf{A}_s)^\top + \mathbf{L}_b \mathbf{R}_v \mathbf{L}_b^\top. \end{aligned} \quad (32)$$

This square-sum form, rather than an independent-noise Joseph formula on the terminal state, is the PSD oracle.

4 Direction-A makes the dynamic builder structured

The apparent obstacle is the family $\mathbf{Q}_{ij}^W$. A generic implementation could materialize b^2 vectors of width d , reproducing the expensive dynamic builder that made the earlier ExactKLA kernel unattractive. Equations (8) and (10) remove the need for dense covariance propagation inside the panel.

For mode m ,

$$\mathbf{Q}_i^{(m)} = \tau_{i,m} \mathbf{I}_{d_m}, \quad \tau_{1,m} = 0, \quad \tau_{i,m} = \rho_{i,m}^2 \tau_{i-1,m} + \nu_{i,m}. \quad (33)$$

Here $\rho_{i,m}$ means the effective radius after applying the reset override, so it is zero at a hard reset. Equivalently, equation (33) is the blockwise form of equation (15) using the effective $\mathbf{D}_i$. For each source observation j , initialize

$$\mathbf{u}_{j,j}^{(m)} = \tau_{j,m} \mathbf{k}_j^{(m)} \quad (34)$$

and propagate forward without a ratio,

$$\mathbf{u}_{i,j}^{(m)} = \mathbf{D}_i^{(m)} \mathbf{u}_{i-1,j}^{(m)}, \quad i > j. \quad (35)$$

Then

$$[\mathbf{\Sigma}_W]_{ij} = \sum_m \mathbf{k}_i^{(m)\top} \mathbf{u}_{i,j}^{(m)}, \quad (36)$$

$$[\mathbf{H}_W]_{ij} = \sum_m \mathbf{q}_i^{(m)\top} \mathbf{u}_{i,j}^{(m)}, \quad (37)$$

$$[\mathbf{C}_W]_{:,j} = \mathbf{u}_{b,j}. \quad (38)$$

The contribution of the dense incoming covariance is computed separately. Form

$$\mathbf{U} = \mathbf{P}^{-\top} \widetilde{\mathbf{K}}. \quad (39)$$

Then

$$\mathbf{G} = \mathbf{R}_v + \widetilde{\mathbf{K}}^\top \mathbf{U} + \boldsymbol{\Sigma}_W, \quad (40)$$

$$\mathbf{H} = \text{tril}(\widetilde{\mathbf{Q}}^\top \mathbf{U} + \mathbf{H}_W), \quad (41)$$

$$\mathbf{C}_b = \boldsymbol{\Phi}_b \mathbf{U} + \mathbf{C}_W. \quad (42)$$

The factor and all right-hand sides can be expressed as one combined solve,

$$[\mathbf{Y}_H \mid \mathbf{Y}_C \mid \mathbf{Z}] = \mathbf{J}^{-1}[\mathbf{H}^\top \mid \mathbf{C}_b^\top \mid \boldsymbol{\Delta}], \quad \mathbf{A}_{\text{out}} = \text{tril}(\mathbf{Y}_H^\top), \quad \mathbf{B}_b = \mathbf{Y}_C^\top. \quad (43)$$

Its right-hand-side width is $b + d + d_v$. At $b = 32$ and $d = d_v = 128$, the solve is 32×288 . The delayed-noise exact-static path solved only $d + d_v = 256$ right-hand sides; its solver timing is therefore not a matched proxy for the dynamic algorithm.

The reference implementation may materialize $\mathbf{u}_{i,j}$ for clarity. A measured kernel must not store a $b \times b \times d$ tensor in global memory. It should propagate a tiled set of live source vectors, accumulate the two $b \times b$ lower triangles, and retain only the $d \times b$ terminal columns.

Crucially, $\boldsymbol{\Sigma}_W$, $\mathbf{H}_W$, and $\mathbf{C}_W$ depend only on $(\mathbf{D}, \boldsymbol{\Omega}, \mathbf{k}, \mathbf{q})$, not on the recurrent dense $(\mathbf{S}, \mathbf{P})$ boundary. One implementation arm may therefore construct the dynamic corrections for all sequence panels in parallel before the dependent dense-boundary loop. The alternative is to fuse their construction into each resident panel. The first arm spends global traffic on packed triangles and terminal columns; the second spends registers and recomputation. Both implement the same equations and must be compared as complete dependent recurrences. A candidate GPU mapping assigns one warp or cooperative tile to each source j , retains one live d -vector, advances it over $i = j:b$, emits packed lower-triangular entries of $\boldsymbol{\Sigma}_W$ and $\mathbf{H}_W$, and writes $\mathbf{C}_{W,:,j}$ only at $i = b$. Radius/angle parameterization is converted to $\rho \cos \theta$ and $\rho \sin \theta$ before this inner source loop; it may not execute a transcendental operation for every (i, j, m) triple.

For the production training shape $B = 4, H = 16, T = 4096, b = 32, d = 128$ and 64 mode variances, a fully materialized FP32 prepass stores, per panel-head, two packed 528-entry triangles, one 128×32 terminal matrix, and 64 variance scalars. This is 20,864 bytes per panel-head and exactly 163 MiB over 8,192 panel-heads. This is only the dynamic-prepass payload, not the total layer workspace. It is transient and preallocated; backward recomputes it rather than checkpointing another copy. The dependent state/factor arenas and backward checkpoint/replay storage require a new complete liveness witness. The existing 248.03125 MiB core witness belongs to the rejected static schedule and cannot be reused as the new total. End-to-end timing includes both the prepass writes and the dependent consumer reads. This schedule targets training and prefill. Token-at-a-time decode remains the $b = 1$ recurrence unless a real token block is available.

4.1 Cost

For one panel, excluding value-width products, the leading work is

Region	Arithmetic
transition prefixes and mode variances	$O(bd)$
dense $\mathbf{P}^{-\top} \widetilde{\mathbf{K}}$ and boundary downdate	$O(bd^2)$
static and dynamic Gram/read systems	$O(b^2d)$
factorization and address-side triangular solves	$O(b^3 + b^2d)$
value solve, outputs, and mean commit	$O(b^2d_v + bdd_v)$

The forward live workspace is $O(d^2 + dd_v + bd + b^2 + bd_v)$ per batch-head item; this excludes batch-head replication and backward checkpoints. The dynamic correction is lower-order than the dense covariance products when $b < d$; in the intended first regime, $b \leq 32 < 128 = d$. This is an arithmetic observation, not a speed result: synchronization, factor latency, register pressure, occupancy, and the number of dependent panels must be measured on H200.

5 Exactness and causality

Proposition 1 (panel equivalence). Under equations (2)–(3), exact arithmetic evaluation of equations (11)–(28) gives the same outputs, terminal posterior mean, and terminal posterior covariance as the tokenwise recursion (5)–(7), for every panel width and every positive observation-noise variance.

Reason. The panel equations construct the joint Gaussian distribution of the panel states and observations. Conditioning this joint distribution on the first i observations gives the same conditional distribution as applying i scalar Gaussian conditioning steps. The leading block of a lower Cholesky factor is the Cholesky factor of the corresponding leading principal block, so equation (25) selects exactly those prefix conditionals. The terminal Schur complement gives equations (27) and (28). A complete algebraic proof appears in Appendix A.

Proposition 2 (causality). Changing any future $\mathbf{k}_t, \mathbf{v}_t, \mathbf{q}_t, \mathbf{D}_t, \mathbf{\Omega}_t$, or r_t with $t > i$ cannot change output $\mathbf{o}_i$.

Reason. The first i rows and columns of $\mathbf{G}$ and $\mathbf{J}$ use only tokens 1: i ; the first i entries of $\mathbf{Z}$ solve the corresponding leading triangular system. Row i of $\text{tril}(\mathbf{H}\mathbf{J}^{-\top})$ has support only on columns 1: i . The full-panel boundary becomes visible only to the next panel.

6 Where approximation is allowed

No statistical approximation is required by panelization itself. If the exact structured builder misses the end-to-end speed target, approximation is confined to the within-panel process-noise kernel.

Let $\mathcal{A}$ be a fixed union of complete rotational modes and decompose

$$\mathbf{Q}_{ij}^W = \mathbf{Q}_{ij}^{W,\mathcal{A}} + \mathbf{Q}_{ij}^{W,\overline{\mathcal{A}}}. \quad (44)$$

Use $\mathbf{Q}^{W,\mathcal{A}}$ consistently in all three inferential objects:

$$\mathbf{\Sigma}_W, \quad \mathbf{H}_W, \quad \mathbf{C}_W, \quad (45)$$

while retaining the full $\mathbf{Q}_b$ in the unconditional terminal covariance. This defines a coherent surrogate: active-mode noise occurs at its original token and is inferred, whereas inactive-mode noise is injected independently at the boundary. The family is nested:

$$d_{\text{active}} = 0 \quad \Longleftrightarrow \quad \text{delayed-noise exact-static ablation}, \quad (46)$$

$$d_{\text{active}} = d \quad \Longleftrightarrow \quad \text{exact dynamic conditioning}. \quad (47)$$

The active set must be fixed before timing and training; selecting it using future panel values would be noncausal. A temporal approximation is also possible, but an arbitrary banding of a

covariance matrix need not remain positive semidefinite. It must therefore be derived from an explicit truncated-noise generative model rather than imposed entrywise.

The following shortcuts are not allowed:

1. adding only $\text{diag}(\Sigma_W)$ without matching state–observation terms;
2. adding C_W without the corresponding innovation covariance;
3. correcting terminal conditioning while leaving causal reads under a different model;
4. moving all dynamics to the panel boundary; or
5. claiming that the delayed-noise covariance is calibrated to the tokenwise process.

7 Why the delayed-noise ablation fails

The exact-static ablation makes the replacement

$$\Sigma_W \leftarrow \mathbf{0}, \quad H_W \leftarrow \mathbf{0}, \quad C_W \leftarrow \mathbf{0}, \quad (48)$$

but still adds Q_b at publication. It preserves the open-loop amount of terminal process uncertainty, yet discards the fact that observations reveal part of that uncertainty. No later boundary addition can reconstruct the missing conditional mean.

The sealed FP64 decision used widths 1, 2, 4, 8, 16, 32, twenty held-out seeds, nine regimes, and $T = 256$. The key failures were

Regime and width	output median	output p95	final-mean median
near-zero transition, $b = 2$	0.6972	0.7356	1.0000
aggressive decay, $b = 2$	0.0069	0.0090	0.3707
near-zero transition, $b = 8$	0.9378	0.9502	1.0000

For the near-zero case at $b = 2$, the sealed per-position output medians were 0.00057 at position one and 0.999996 at position two—exactly where equation (48) removes the newly injected state from conditioning. Independent token-loop and panel implementations agree on the surrogate, and deterministic panels with $\Omega_{2:b} = 0$ recover tokenwise ExactKLA. These controls localize the discrepancy to the intended approximation. This historical decision is bound to manifest `f21efd8a7725...5923215`, source seal `603ae064682a...84a87b`, result body `bd255dbaa235...84f111`, and repository commit `923695601543711c3a7afbcde256ea2435b6dd9d`. The sealed source bytes still match, but the repository HEAD has since advanced; the current verifier therefore rejects the environment binding unless run from the sealed checkout. The table reports that historical verified decision rather than claiming a fresh result at the current HEAD.

8 Implementation and evidence plan

All development occurs under a new sandbox. Existing production mixers and the sealed delayed-noise evidence are read-only.

8.1 Stage 1: independent FP64 references

Implement three paths:

1. the dense tokenwise recursion in equations (5)–(7);
2. a source-loading joint-Gaussian oracle that does not use filtering or the panel identities; and
3. the structured panel equations, first with explicit $\mathbf{u}_{i,j}$ and then with the intended streaming builder.

Compare all outputs, every true panel boundary, and the final state for widths 1, 2, 3, 4, 8, 16, 32; angles 0, $\pi/2$, π , random angles, and time-varying angles; radii 0, 0.05, 0.999 and random contractions; zero, small, and high process noise; repeated, collinear, and zero keys; partial tails; arbitrary dense incoming covariance; and interior resets. Required controls are

1. $b = 1$ equals tokenwise filtering;
2. $\nu_{2,b} = 0$ equals exact-static conditioning;
3. the pair-tied $\theta = 0$ case equals the frozen diagonal-transition direct-conditioning reference; and
4. independent mutations deleting Σ_W , $\mathbf{H}_W$, or $\mathbf{C}_W$ are each detected by a named failing test.

8.2 Stage 2: causality and gradients

Mutate every future input independently and prove that earlier outputs are unchanged. Poison future triangular storage and inactive tail lanes. Run FP64 `gradcheck` and independent vector–Jacobian comparisons for

$$\mathbf{q}, \mathbf{k}, \mathbf{v}, r, \rho, \theta, \nu, \mathbf{S}_0, \mathbf{P}_0, \quad (49)$$

reporting every input separately. Losses must touch current outputs, terminal mean, and terminal covariance. Symmetric $\mathbf{P}_0$ cotangents are part of the contract.

8.3 Stage 3: FP32 numerical gate

Use BF16 layer inputs promoted to FP32 state arithmetic. Begin with IEEE FP32 and TF32 disabled. Compare full $T = 4096$ dependent sequences and gradients with FP64. Immediately before Cholesky, form $\mathbf{G} = \frac{1}{2}(\mathbf{G}_{\text{raw}} + \mathbf{G}_{\text{raw}}^\top)$ from the fully assembled static and dynamic terms. Apart from this Gram symmetrization, symmetrize only complete physical covariance matrices; do not silently jitter, clamp, or project eigenvalues. Any retry or fallback is part of the declared algorithm and final timing. Causal prefix factors use an unpivoted chronological Cholesky. A panel-global jitter, pivot, or precision decision triggered by a future row may not revise an earlier output; any causal-output retry must itself be prefix-local, while a terminal-only retry may affect only the boundary published after the panel. For a tensor with n entries, require

$$\|x - x_{\text{ref}}\|_F \leq \text{atol}\sqrt{n} + \text{rtol}\|x_{\text{ref}}\|_F. \quad (50)$$

The proposed limits, to be sealed in a new manifest before any candidate results are inspected, are $(2 \times 10^{-5}, 5 \times 10^{-4})$ for random outputs/means, $(5 \times 10^{-5}, 10^{-3})$ for random covariances, $(2 \times 10^{-4}, 5 \times 10^{-3})$ for stress outputs/means, $(5 \times 10^{-4}, 10^{-2})$ for stress covariances, and $(10^{-4}, 10^{-2})$ for each VJP. Every boundary must satisfy $\lambda_{\min}(\mathbf{P}) \geq -10^{-5} \max(1, \lambda_{\max}(\mathbf{P}))$, with normalized pre-symmetrization antisymmetric residual at most 10^{-4} . Any nonfinite value or Cholesky failure rejects the arm.

8.4 Stage 4: H200 feasibility attribution

At production geometry $B = 4, T = 4096, D = 2048, H = 16, d = d_v = 128$, compare five named arms: the complete KDA layer; the runnable exact direct-conditioning fallback; the fastest same-source delayed-noise solve with 256 right-hand sides; a schedule-matched delayed-noise control padded to 288 right-hand sides; and structured-exact dynamic conditioning with 288 right-hand sides. The existing G5 work is a scheduling ancestor and a $\theta = 0$ semantic/resource control, not a headline speed baseline until its own full qualification is complete. For context only, the old runnable generic exact layer measured 77.3749 ms versus 2.63338 ms for KDA in a non-final screen; its materializing builder/factor/solve/epilogue medians were 54.937/7.600/6.980/4.659 ms, while a prepared two-CTA 32-row solve alone measured 1.3401 ms. None of those historical component numbers is new candidate evidence. The historical fallback uses the older transition model. Before it can define a speed ratio, add or rerun a correctness-qualified exact fallback on the identical rotational inputs and layer peripherals; otherwise report the historical row only as context.

Attribute

1. transport and mode-variance prefixes;
2. dense $P^{-}\widetilde{K}$;
3. static Gram/read construction;
4. the structured dynamic correction;
5. factorization and combined triangular solve;
6. causal output, mean commit, and dense covariance downdate; and
7. the complete dependent panel loop.

Profile $b \in \{4, 8, 16, 32\}$, mode counts $M \in \{8, 16, 64\}$, $T \in \{128, 512, 1024, 4096\}$, and batch-head counts 1, 6, 16, 64. The primary decision point is $B = 4, H = 16, T = 4096, b = 32, d = d_v = 128$. A component sum is not a performance claim. The decisive call is one literal recurrence in which every panel consumes the actual dense boundary emitted by its predecessor. Report compiler resources, spills, allocator traffic, launch count, peak memory, and achieved throughput. A persistent outer scheduling tile may contain several statistical panels, but it may not change equations (25)–(28). Compare an all-panels-parallel dynamic prepass with inline construction. Attribute its packed global storage and traffic explicitly; a fast prepass timed without its consumption by the boundary recurrence is only a component result. The inline arm must report zero unintended stack, local, spill, or global scratch and capacity for at least 64 active programs or clusters at batch-head 64; the prepass arm targets at least two resident CTAs per SM. Both must account for the raw (ρ, θ) to rotation-coefficient transform. Phase-dead storage may be aliased only after a liveness audit proves the alias.

8.5 Stage 5: exact versus controlled approximation

First benchmark the structured-exact builder. Only if it misses the frozen complete-layer target should the mode-selective family in equation (45) be evaluated. Freeze the active-mode sets and quality thresholds before reading held-out results. A nontrivial approximation must pass sequence-level output, boundary-mean, true-MSE, and language-model quality gates; one-panel covariance agreement is insufficient. The already rejected $d_{\text{active}} = 0$ endpoint cannot be reconsidered under a new name.

8.6 Stage 6: production admission

Use a source-pinned, allocation-free, same-process interleaved AB/BA benchmark against the real incumbent ExactKLA path and the complete KDA layer. Median and minimum ratios must agree on the decision. Forward and forward-backward are separate gates, and mandatory predicates, tails, symmetrization, empty fallback dispatches, and layer peripherals remain inside the timed call. Use 15 AB/BA warm-up pairs followed by 40 measured pairs over identical preallocated inputs, with a KDA sentinel every five pairs; reject more than 5% first-to-last-quartile sentinel drift. The following are new proposed admission gates for structured dynamic panels. They are not inherited from the historical ExactKLA contract and must be sealed before timing. The complete forward and forward-backward paths must each be at most $3.0\times$ KDA by both median/median and minimum/minimum ratios, at least $4.0\times$ faster than the runnable exact fallback, peak at most $2.0\times$ KDA allocated memory and below the fallback, and perform no steady-state allocator calls. “Allocation-free” means no calls after setup with the complete workspace declared by the new forward/backward liveness witness; this includes, but is not limited to, the 163 MiB prepass payload. A prepass design may use at most two forward core launches (prepass plus dependent core); an inline persistent design may use one. Backward recomputes dynamic corrections and has its own complete timing and memory gate. Only after correctness, gradients, numerical behavior, resources, memory, and speed pass may the candidate be copied into `lit_gpt/`.

9 Limits

The derivation does not prove that a fast kernel exists. It removes a statistical approximation by exploiting model structure, but still retains dense $O(Td^2)$ covariance arithmetic and a dependent boundary recurrence. Small-panel Cholesky latency, the $b + d + d_v$ right-hand-side solve, limited batch-head parallelism, prepass traffic, rotation-coefficient preparation, and backward checkpointing may dominate despite favorable operation counts. Conversely, the old generic dynamic builder timings do not directly reject this formulation because they materialized substantially more transition-conditioned data and targeted larger dense solves. Only the complete dependent measurement can decide.

The block-isotropic restriction applies to process noise, not to the posterior covariance. A general dense $\mathbf{\Omega}_t$ destroys the scalar mode recurrence in equation (33); it remains mathematically compatible with the panel equations but not with the claimed structured-builder cost. Likewise, state-dependent transitions that are unknown before the scan fall outside this formulation.

A Proof of panel equivalence

For one value coordinate, stack the observations $\mathbf{y} = [v_1, \dots, v_b]^\top$. Equations (14) and (3) give

$$v_i = \tilde{\mathbf{k}}_i^\top \mathbf{x}^\star + \mathbf{k}_i^\top \boldsymbol{\xi}_i + e_i. \quad (51)$$

The conditional mean of $\mathbf{y}$ before observing the panel is $\tilde{\mathbf{K}}^\top \mathbf{X}^-$. Independence of the entry-state error $\mathbf{x}^\star - \mathbf{X}^-$, $\mathbf{w}_{2:b}$, and $\mathbf{e}_{1:b}$ gives

$$\text{Cov}(v_i, v_j) = \tilde{\mathbf{k}}_i^\top \mathbf{P}^- \tilde{\mathbf{k}}_j + \mathbf{k}_i^\top \mathbf{Q}_{ij}^W \mathbf{k}_j + \mathbf{1}_{\{i=j\}} r_i, \quad (52)$$

for $i \geq j$, with the other triangle obtained by symmetry. This is equation (21). For $j \leq i$,

$$\text{Cov}(\mathbf{s}_i^*, v_j) = \text{Cov}(\Phi_i \mathbf{x}^* + \xi_i, \tilde{\mathbf{k}}_j^\top \mathbf{x}^* + \mathbf{k}_j^\top \xi_j + e_j) \quad (53)$$

$$= \Phi_i \mathbf{P}^- \tilde{\mathbf{k}}_j + \mathbf{Q}_{ij}^W \mathbf{k}_j, \quad (54)$$

which is equation (23).

For any jointly Gaussian variables a and y ,

$$\mathbb{E}[a \mid y] = \mathbb{E}[a] + \text{Cov}(a, y) \text{Cov}(y)^{-1} (y - \mathbb{E}[y]), \quad (55)$$

and

$$\text{Cov}(a \mid y) = \text{Cov}(a) - \text{Cov}(a, y) \text{Cov}(y)^{-1} \text{Cov}(y, a). \quad (56)$$

Taking $a = \mathbf{s}_b^*$, using $\mathbf{G}^{-1} = \mathbf{J}^{-\top} \mathbf{J}^{-1}$, and defining $\mathbf{B}_b = \mathbf{C}_b \mathbf{J}^{-\top}$ yields equations (27)–(28).

For output i , apply equation (55) to $a = \mathbf{q}_i^\top \mathbf{s}_i^*$ and the prefix $\mathbf{y}_{1:i}$. The leading principal block $\mathbf{J}_{1:i, 1:i}$ factors $\mathbf{G}_{1:i, 1:i}$. The first i entries of $\mathbf{Z}$ equal $\mathbf{J}_{1:i, 1:i}^{-1} \Delta_{1:i}$, and the first i entries of row i of $\mathbf{H} \mathbf{J}^{-\top}$ equal $\mathbf{H}_{i, 1:i} \mathbf{J}_{1:i, 1:i}^{-\top}$. Equation (25) follows.

Sequential Kalman filtering and joint Gaussian conditioning are two factorizations of the same conditional distribution. Applying the argument panel by panel proves equality for a complete sequence, including partial final panels.

B The exact-static discrepancy

Let $\mathbf{L}_X = \mathbf{P}^- \tilde{\mathbf{K}} (\mathbf{R}_v + \tilde{\mathbf{K}}^\top \mathbf{P}^- \tilde{\mathbf{K}})^{-1}$ be the block gain used by the delayed-noise surrogate, and set

$$\mathbf{L}_b = \Phi_b \mathbf{L}_X, \quad \mathbf{A}_b = \Phi_b (\mathbf{I} - \mathbf{L}_X \tilde{\mathbf{K}}^\top). \quad (57)$$

If $\mathbf{E}^-$ is the actual incoming estimation-error second moment and $\mathbf{P}^-$ is the covariance published by the surrogate, define $\Delta_P^- = \mathbf{E}^- - \mathbf{P}^-$. The true terminal MSE of the surrogate differs from its published covariance by

$$\mathbf{E}_b - \mathbf{P}_b^{\text{static}} = \mathbf{A}_b \Delta_P^- \mathbf{A}_b^\top - \mathbf{L}_b \mathbf{C}_W^\top - \mathbf{C}_W \mathbf{L}_b^\top + \mathbf{L}_b \Sigma_W \mathbf{L}_b^\top. \quad (58)$$

The difference need not be positive semidefinite: intermediate process noise both contaminates the observations and contains information about the terminal state. This identity explains why adding the correct open-loop $\mathbf{Q}_b$ cannot calibrate a gain that ignored how the noise entered the observations.